Zhiyu Wang

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Google Scholar LinkedIn ORCID

Agents | LLMs | AI4Bio/Health | Geometric DL



Education

University of Cambridge — Cambridge, UK
Advanced Computer Science, Master of Philosophy (Distinction, Rank: 8/60) — Oct 2024 — Jul 2025

- Thesis awarded the Highly Commended MPhil Project Prize 2024–2025 by the Department of Computer Science and Technology

University College London — London, UK
Computer Science, Bachelor of Science (First Class Honors) — Sep 2021 — Jun 2024

Research Experience

Auto-Research for the Virtual Cell Challenge | Sapient Intelligence & SJTU — Remote
Project lead | *Collaborator: Prof. Liang Hong* — Aug 2026 — Present

- Leading a joint project with Shanghai Jiao Tong University (SJTU) that applies Praxist, our AI-scientist framework, to zero-shot prediction of CRISPRi knockdown responses in the **Virtual Cell Challenge 2026**; ranked 207/1,490 (top 13.9%) on the preliminary leaderboard.

Bayesian Optimization for Auto-Research Systems | Mila – Quebec AI Institute — Remote
Collaborator | *Supervisor: Prof. Jian Tang* — Apr 2026 — Present

- Integrated Bayesian optimization into agentic evolution: a Gaussian process predicts gains; Thompson sampling picks the next experiment.
- Achieved the best average rank on 15 AlphaEvolve tasks and the top gold-medal rate (45%) on MLE-bench Lite.

Protein Substructure Alignment via Optimal Transport | University of Cambridge & SJTU — Cambridge, UK
Project lead | *Supervisor: Prof. Pietro Liò & Prof. Liang Hong* — Apr 2025 — Sep 2025

- Developed PLASMA, a plug-and-play optimal-transport module for local substructure alignment on any protein language model. (**ICLR26**)
- In detecting shared substructures, PLASMA reached ROC-AUC 0.96–0.99 vs. 0.81–0.93 for the widely used global aligner TM-align at $\sim$50$\times$ its speed, and outperformed the state-of-the-art local method EBA at $\sim$3$\times$ its speed, with much cleaner alignments.
- Showed with PyMOL case studies that the aligned substructures are structurally similar, not artifacts of sequence similarity.

Hierarchical Protein Representation Learning | University of Cambridge — Cambridge, UK
Master's thesis | *Supervisor: Prof. Pietro Liò* — Oct 2024 — Jul 2025

- Built a complete SE(3)-equivariant framework that adds a secondary-structure level to protein representations via a hierarchical hypergraph, allowing a coarser-grained representation that gives better global understanding. (**Invited Talk, Cambridge**) (**arXiv**)
- Leveraged this new structural hierarchy to achieve the best SCOP fold classification accuracy (43.3% vs. 38.4% for the same network without it; 40.1% best baseline), confirming the improved global understanding of protein structure.

Multi-Omics Data Integration with GNNs | University of Cambridge — Cambridge, UK
Project lead | *Supervisor: Prof. Pietro Liò* — Feb 2025 — Sep 2025

- Built MoRE-GNN, a heterogeneous graph autoencoder learning relational graphs from single-cell RNA, protein and ATAC data; outperforms MOJITOO on strongly correlated CITE-seq data (BM-CITE ARI 0.892 vs 0.868) and enables cross-modal prediction. (**arXiv**)

Graph-based Pain Estimation from Facial Expressions | University of Cambridge — Cambridge, UK
Project lead | *Supervisor: Prof. Hatice Gunes* — Oct 2024 — Jan 2025

- Proposed GraphAU-Pain, combining a ResNet-50 backbone with a facial action unit (AU) graph for interpretable pain estimation; transfer learning for domain shift and missing labels raised average AU F1 on UNBC from 20% to 63%. (**IJCAI25 Workshop**) (**IEEE CEM**)

Readmission Prediction with Remote Patient Monitoring Data | University College London — London, UK
Undergraduate thesis | *Supervisor: Prof. Ivana Drobnjak* — May 2023 — Jun 2024

- Built readmission-prediction models (ROC-AUC 0.79±0.03) from remote patient monitoring data with UCL Hospital clinicians, using SHAP to show which clinical features drive each prediction and patient-independent bootstrapping to cope with only 55 patients.

Industrial Experience

Autonomous Research Agents | Sapient Intelligence — Beijing, China
ML Research Scientist — Sep 2025 — Present

- Co-first author of **Praxist**, an autonomous AI-scientist framework with DAG-based execution lineage; on the 75-task MLE-bench, it won 60 medals (49 gold) vs. 55 (34 gold) for Claude Code with Opus, at about 1/12 of the cost.
- Showcased Praxist at **AI4 2026** through live booth demos and conversations with prospective customers and collaborators.

Healthcare LLM Knowledge Distillation | iFLYTEK Healthcare — Hefei, China
Intern ML Researcher — Jul 2024 — Sep 2024

- Distilled a teacher LLM into synthetic medical instruction data to fine-tune the model behind iFLYTEK's **Xiaoyi** AI health-assistant app, raising instruction-following accuracy from 34% to 82%.

RAG-Based Financial Intelligence System | Luojin Data Information — Shanghai, China
Intern ML Engineer — May 2023 — Aug 2023

- Built a RAG pipeline (`MongoDB`, `LangChain`) for financial report classification with 97% entity matching.

Homepage Navigation Assistant for Hospitals | UCL & National Health Service — London, UK
Industrial Collaboration Program — Sep 2022 — May 2023

- Built a system that crawls a hospital's website and auto-generates an assistant that answers visitor questions and guides them to the right pages, cutting setup from weeks of manual development to 30 minutes; demoed at Great Ormond Street Hospital (**media coverage**).

Publications

Conference, Journal & Workshop Papers — ** equal contribution*

1. **"Fast and Interpretable Protein Substructure Alignment via Optimal Transport."**
   Zhiyu Wang*, Bingxin Zhou*, Jing Wang, Yang Tan, Weishu Zhao, Pietro Liò, Liang Hong. **ICLR 2026**.
2. **"Data-Efficient AU-Graph Transfer Learning for Consumer-Camera Pain-Aware Sensing."**
   Zhiyu Wang, Han Wei Wang, Hangqian Li, Jun Jiang, Hatice Gunes, Yang Liu. **IEEE Consumer Electronics Magazine 2026**.
3. **"GraphAU-Pain: Graph-based Action Unit Representation for Pain Intensity Estimation."**
   Zhiyu Wang, Yang Liu, Hatice Gunes. **IJCAI 2025 MiGA Workshop**.

Preprints

4. **"Praxist: From Experimental Artifacts to Solution Lineages."**
   Jin Li*, Ahmed Murtadha*, **Zhiyu Wang***, Qiwen Chen*, William Chen*, Yifei Wu*, Guan Wang*, et al. **arXiv 2026**.
5. **"Topotein: Topological Deep Learning for Protein Representation Learning."**
   Zhiyu Wang, Arian Jamasb, Mustafa Hajij, Alex Morehead, Luke Braithwaite, Pietro Liò. **arXiv 2025**.
6. **"MoRE-GNN: Multi-omics Data Integration with a Heterogeneous Graph Autoencoder."**
   Zhiyu Wang*, Sonia Koszut*, Pietro Liò, Francesco Ceccarelli. **arXiv 2025**.

Additional Experience

- **Reviewer Experience:** NeurIPS 2026; AI4Science Workshop at NeurIPS 2025 and ICML 2026
- **Teaching Experience:** Programming Tutor at UCL (2022–23)